The Law of Narrative Momentum — Theory

First-Class History, Directional Persistence, and Revision

Matthew Parslow
Independent Researcher

Draft — draft-2026-08-26
26 August 2026

DRAFT — draft-2026-08-26 (26 August 2026). This document is under active revision and is circulated for comment, not as a finished result. Definitions, numbering and claims may change between drafts; cite the version string above if referring to it. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

This paper develops Narrative as a real, measurable, frame-relative quantity rather than as a metaphorical story stored inside the graph it describes. Graphic Equalisation establishes that a containing frame may possess first-class relational state about a lower graph even where that quantity is absent from the lower graph itself. Realised history can therefore exist as ordinary state in the representing frame.

Narrative is defined as a discrete frame-level measure of realised-history structure that remains operative in the present representation of a graph. The Narrative dimension need not exist in every frame; where it is absent, Narrative is undefined rather than numerically zero. Where history-bearing structure persists across collapse, it enters the next antecedent configuration and thereby biases subsequent reachable geometry toward continuations compatible with that realised structure. Momentum is consequently not introduced by an unexplained grounding arrow from Narrative to a second ontological object. It is the directional dynamical consequence of operative Narrative in a geometry that supplies direction.

The resulting law is structurally adjacent to established work on hysteresis, path dependence, belief revision, predictive coding, motor schemas, and institutional lock-in, but is not identical to any one of them. The contribution here is the common frame-relative formulation: realised history becomes first-class state, present state carries only the history that remains causally operative, and directional persistence follows from its presence in the next possibility relation.

1 Dependencies and scope

Reading for the claim. As in the foundational paper, the components used here are individually established and are cited as such. The claim is the arrangement, and it is not reachable by evaluating the components separately, since those are stipulated to be known. The composition's own falsifiability commitment is stated in the foundational paper and is inherited here.

Graphic Equalisation supplies deterministic realised transition

$$G_t \xrightarrow{\mathcal{C}_t} G_{t+1},$$

and the Seldon-manifold theory supplies counterfactual reachable geometry

$$\mathcal{S}_{h,\rho}(\mathcal{F}_t).$$

This paper adds a measurable higher-frame quantity carrying realised history. It does not require cognition, self-models, language, or deliberative agency.

The nearest established structures are history-dependent dynamics and hysteresis, in which present response depends on retained information about past input [2]; economic and institutional path dependence, in which earlier events alter later reachable trajectories [3–5]; formal belief revision, in which changing one proposition may require revision of a larger dependency structure [6]; and predictive-processing models in which present expectations shape interpretation of subsequent input [7]. These provide prior art for particular behaviours. The present claim is the cross-domain identity of the underlying frame-relative structure.

2 Narrative as a first-class dimension

Let W individuate graph G.

Definition 2.1 (Narrative dimension)**.** *The Narrative dimension $D_{\mathcal{N},W:G}$ exists in frame W when W represents a history-bearing relation concerning G whose value is causally operative in subsequent transition or reachability structure.*

Before that relation exists,

$$D_{\mathcal{N},W:G} = \text{NULL},$$

and $\mathcal{N}_{W:G,t}$ is undefined. This is not the numerical state $\mathcal{N}_{W:G,t} = 0$. Existence of the dimension precedes value on the dimension.

Definition 2.2 (Narrative)**.** *Where $D_{\mathcal{N},W:G}$ exists,*

$$\boxed{\mathcal{N}_{W:G,t} \in \mathbb{Z}}$$

is the discrete frame-relative measure of realised-history structure concerning G that remains represented and operative in W at realised step t.

The integer is a state index rather than a globally fixed unit. A grounding may map it to arbitrary dimensional scales or dynamically varying quantisation. A value of zero is a legitimate value on an instantiated Narrative dimension; it is not absence of that dimension.

Proposition 2.3 (No emergence gap)**.** *The absence of Narrative as a primitive in $G = (g, r, \Gamma)$ does not make Narrative non-physical or metaphorical. Where the defining relation exists in W, Narrative is ordinary first-class state in W.*

This is a direct use of the higher-frame principle established in Graphic Equalisation: a relation may be absent from the lower graph while being fully real in a frame containing the relation required to define it.

3 Narrative and realised history

Realised history is the represented succession

$$\mathcal{H}_{W:G,\leq t} = G_0 \xrightarrow{\mathcal{C}_0} G_1 \xrightarrow{\mathcal{C}_1} \cdots \xrightarrow{\mathcal{C}_{t-1}} G_t.$$

Narrative is not necessarily the complete history. It is the currently operative history-bearing structure:

$$\mathcal{H}_{W:G,\leq t} \longrightarrow \mathcal{N}_{W:G,t}.$$

Proposition 3.1 (Historical sufficiency of present structure)**.** *Past events affect future dynamics only insofar as their consequences remain represented in present graph or boundary structure.*

This proposition is closely related to the state-variable treatment of systems with memory: the past influences future behaviour through whichever memory variables remain operative in the present representation. Hysteresis theory makes this explicit in systems whose present response retains selective information about past extrema or branches [1, 2]. The present theory generalises the statement across frames and graph types.

Narrative is therefore not "everything that happened." It is the part of realised history that has not disappeared dynamically.

4 Direction and momentum

Narrative magnitude alone is not a vector. Direction exists only where the relevant frame also instantiates a directional dimension or ordering over continuation.

Let $\hat{\mathbf{d}}_{W:G,t}$ denote a locally represented direction associated with the history-bearing relation. In a grounding where scalar Narrative magnitude and directional coordinates are commensurable, a convenient coordinate representation is

$$\boxed{\mathbf{p}_{W:G,t} = \phi_W(\mathcal{N}_{W:G,t})\,\hat{\mathbf{d}}_{W:G,t},}$$

where ϕ_W supplies the frame-specific scale. The special case $\phi_W(n) = n$ gives $\mathbf{p} = \mathcal{N}\hat{\mathbf{d}}$.

Definition 4.1 (Narrative momentum)**.** *Narrative momentum is the directional persistence in reachable future geometry induced by operative Narrative when a directional dimension exists in the frame.*

Momentum is therefore derived from first-class Narrative plus represented geometry; it is not posited as a second unexplained substance. Classical momentum is one exact grounding of this structural form, not the definition from which the abstract quantity is borrowed.

5 Momentum is present structure

Before the law is stated, the notion of momentum it uses must be made explicit, because the word invites a reading this theory does not take.

Proposition 5.1 (Momentum is not history carried through time)**.** *Momentum is a* currently instantiated structural consequence *of prior change, which constrains what can happen next:*

$$\boxed{\textit{prior change} \;\rightarrow\; \textit{present momentum} \;\rightarrow\; \textit{determinable continuation.}}$$

The past need not continue to exist for momentum to be real. What is real is the structure prior change left behind, instantiated now.

This is Graphic Equalisation's history principle applied to a specific quantity: history matters only where it survives as present structure. A momentum term is exactly such surviving structure.

Remark (Narrative momentum is not a borrowed metaphor). Ordinary physical momentum and Narrative momentum have the same structural form under Proposition 5.1: prior deformation leaves present structure that constrains reachable continuation. Narrative momentum therefore requires no defence as an analogy imported from mechanics. The two are instances of one form, differing in which relations carry the surviving structure — and in the case of Narrative, in the fact that those relations are represented rather than material.

Remark (Momentum is the un-rewritten Narrative; to change course is to change it). What a frame carries forward is not a bare delta but its operative Narrative — the surviving history-bearing structure S of Section 5. The Narrative is not pointed at through any indirection; it is the frame's own standing structure, and a delta is a write *to* it — a self-change of S, gated by the container (Graphic Equalisation) — not a separate thing the frame carries. Continuation is then not a repeated action but the absence of one: nothing re-applies S each step, because there is no step — no global tick — for it to be re-applied at (Graphic Equalisation, collapse as a local write). S persists until a rule writes against it, and that write is the only priced operation. This is why the "Nothing pushes" reading (Remark 6) is exact rather than a manner of speaking: persistence is the free default and change the paid exception, because retaining the Narrative is not a write and altering it is. To change course is therefore, literally, to change the Narrative — a write to S — and its price is the alteration cost C_{alt} of Hypothesis 6.1, which in the mechanical grounding is the impulse $J = \Delta\mathbf{p}$ (Remark 10).

It also fixes what the momentum quantity is. Read from the Narrative, it is a difference, Δ, and not a difference per unit time, $\Delta/\Delta T$: the divisor is the inferred time dimension, which is not available in-frame (Graphic Equalisation, the missing denominator). "Velocity" is therefore an observer's cross-frame reading of the Narrative's persisting difference, not the substrate quantity, and the substrate statement of the first law is that the Narrative continues un-rewritten until a write — a change of the Narrative — changes it.

6 The Law of Narrative Momentum

The law is stated first without reference to direction, because Graphic Equalisation defines none. Realised collapse induces an *order* over events; it does not supply a basis, an orientation, or a vector space. Direction is a frame's optional instrumentation (Section 5), and a law that presupposed it would be a claim about representations rather than about reachability. The directional statement is recovered below as the special case for frames that instantiate a directional dimension.

Hypothesis 6.1 (Law of Narrative Momentum, cost form)**.** *Let S be operative history-bearing structure that survives a realised transition and remains materially unchanged across the next one. Then for each admissible continuation $a \in \mathcal{P}(G_{t+1})$, the cost of a is conditioned on S:*

$$C(a \mid S) = \begin{cases} C_0(a) & a \text{ reuses } S, \\ C_0(a) + C_{\text{alt}}(S, a) & a \text{ requires } S \text{ altered or replaced,} \end{cases}$$

where C_{alt} is the cost of the interaction that alters S.

Remark (What is conditioned, and what is not). Γ **decides membership; Narrative decides cost.** Nothing leaves the admissible set because of history. The set is what the admissibility grammar admits, and surviving structure prices the members. This division is what keeps the two quantities separable: an account that let history remove continuations would have Narrative doing Γ's work, and no way to tell which was responsible for an absence.

Remark (Nothing pushes). The hypothesis posits no force, no tendency, and no stored impetus. The whole asymmetry is that *continuing costs nothing and altering costs*: S is already instantiated, so a continuation employing it requires no further expenditure, while one contradicting it must pay C_{alt}. Persistence is what an unaltered structure looks like from the next step, not something acting on the graph.

Remark (Irreversibility is a limit case, not a removal). Where $C_{\text{alt}}(S, a)$ exceeds any expenditure available in the frame, a is unreachable in practice while remaining admissible. From inside the frame this is indistinguishable from inadmissibility; it is different in kind, and conflating them

relocates a budget fact into the grammar. Irreversibility is therefore frame- and budget-relative rather than absolute, consistent with the companion manifold paper's treatment of it as a property of a deformation rather than of an act.

Corollary 6.2 (Material redirection)**.** *Redirection therefore requires paying C_{alt} — an interaction that changes the relations carrying S. It is not achieved by preference over continuations, by representing a different continuation, or by intending one: those leave S instantiated, so the price stands and the cheap continuations remain the ones that reuse it.*

Corollary 6.3 (Directional form)**.** *Where frame W instantiates a directional dimension and $\mathcal{N}_{W:G,t}$ is operative, the cost conditioning of Hypothesis 6.1 projects into that dimension, and subsequent reachable geometry retains a directional component aligned with the continuation the Narrative represents — because the cheapest continuations are those reusing S. This is the statement in directional language, and it holds only where the frame supplies the direction. Two frames may instantiate opposed directional dimensions over the same S and disagree about the sign of the retained component while agreeing on the conditioning that produces it.*

Remark (Why the restriction form is the substrate-level statement). Under the directional statement alone, a frame with no directional dimension makes the law vacuous, which would be a strange property for a claim about what is reachable. The cost form applies wherever a possibility set exists, requires no basis, and is invariant under the choice of one — a gauge condition of the same kind the companion construction paper imposes on relative weighting. The measured groundings below are of this form: a worn route does not possess a direction, it possesses differential cost, and two travellers moving oppositely along it experience opposite directions from identical structure.

The key recursion is

$$\boxed{\text{history survives as structure} \Rightarrow \text{history enters the next possibility relation.}}$$

The law itself is deterministic. For empirical work over repeated cases where the observer does not represent every determining dimension *and instantiates a directional dimension*, so that Corollary 6.3 applies, an ensemble statistic such as

$$\frac{1}{m}\sum_{i=1}^{m} \mathbf{p}_{t+1}^{(i)} \cdot \widehat{\mathbf{p}}_{t}^{(i)} > 0$$

may test directional persistence across cases. The statistic measures the observer's population-level evidence; it does not add stochasticity to the Graph Law.

7 Narrative revision

Definition 7.1 (Narrative revision)**.** *A material Narrative revision is a realised change*

$$\Delta\mathcal{N} \neq 0$$

large enough to alter future distinctions or direction at the operative frame and resolution.

A proposition may change while Narrative does not. Conversely, a small local change may induce a large Narrative revision when it alters a highly connected history-bearing relation.

Formal belief-revision theory already distinguishes local proposition change from revision of a larger belief set and studies the conditions under which contraction or revision preserves as much prior structure as possible [6]. The present framework differs by treating the relevant dependency structure as dynamical graph state rather than purely propositional closure, but the structural predecessor is direct.

8 Structural lock-in

Hypothesis 8.1 (Structural lock-in)**.** *For fixed frame and revision class, the minimum intervention required for material Narrative revision tends to increase with the amount of present relational structure that must be modified, bypassed, or rendered irrelevant.*

If $K_R(\mathcal{N})$ is a discrete measure of Narrative-supporting structure, an appropriate monotone grounding predicts that increasing K_R does not decrease minimum revision cost after relevant capacity and geometry are controlled.

This generalises familiar forms of lock-in. In technology and economics, increasing returns can make an historically selected path progressively harder to leave [3, 4]. In institutional analysis, timing, sequence, and increasing returns can similarly make established arrangements difficult to reverse [5]. These are groundings of the structural-lock-in hypothesis, not its proof.

9 History-sensitive projection

Let H_t be dynamically relevant history and $\widehat{H}_t$ represented history.

Hypothesis 9.1 (History-sensitive projection)**.** *Where two represented histories imply different operative present relations despite selected instantaneous observables matching, their projected reachable geometries diverge over some relevant horizon:*

$$\widehat{H}_t \neq H_t \Rightarrow \widehat{\mathcal{S}}_{h,\rho}(\mathcal{F}_t) \neq \mathcal{S}_{h,\rho}(\mathcal{F}_t).$$

The load-bearing clause is "different operative present relations." If the complete causally operative present state is identical, Graphic Equalisation is deterministic and the histories cannot produce different successors merely because they were different histories. History matters only where it survives in state, exactly as Proposition 3.1 requires.

Predictive-coding models provide a cognitive analogue: higher-level predictive state shapes the processing of subsequent input, while residual error updates that state [7]. Motor-control schema theory likewise treats prior movement experience as modifying generalized structures used to parameterise later action [8]. Neither is claimed to be identical to Narrative Momentum; both demonstrate established domain-specific cases in which retained history changes future projection.

10 Mechanical grounding

Classical momentum is a constrained quantitative grounding. In a mechanical frame, realised positional history together with a time embedding determines velocity; mass supplies the established mechanical scale and

$$\mathbf{p} = m\mathbf{v}.$$

The present theory does not redefine classical momentum. It identifies the same abstract ingredients—frame-relative history-bearing state, direction, persistence, and redirection—in a domain where the quantitative law is already exact.

Remark (Under the cost form the correspondence is an identity, not an analogy). Newton's first law is the statement that continuing costs nothing. Altering motion requires impulse, and the impulse needed to bring a body to rest is

$$J = \Delta\mathbf{p} = m\mathbf{v},$$

so *the momentum of a body is exactly the cost of stopping it.* Hypothesis 6.1 is that sentence with the currency left unspecified: C_{alt} in the mechanical grounding is impulse, and mass is the exchange rate converting a desired change of motion into required expenditure.

Remark (Two currencies for one alteration). The same alteration has two established mechanical prices: $m\mathbf{v}$ in impulse and $\frac{1}{2}mv^2$ in energy. They are not proportional, and neither is more correct. This is a concrete instance of the commitment that costs need not share a unit or an origin across frames or classes — a point that otherwise reads as a hedge, and which classical mechanics already demonstrates within a single grounding.

Remark (A grounding, not an analogy). The correspondence should not be described as analogical. Under Axiom −1 an inertial reference frame and a representational frame are the same kind of object — a locus at which distinctions are made, with quantities indexed to it and no view from nowhere available. They differ in how much structure is instantiated, not in kind. A mechanical grounding is therefore an *instance* of the present claims in a frame carrying a great deal of structure, and calling it an analogy would reintroduce the privileged-physical / derivative-representational division that Axiom −1 removes.

What must not be overread is *scope*, which is a different and real constraint: Narrative is not asserted to carry SI momentum units, non-mechanical groundings need not obey conservation of mechanical momentum, and no claim is made that a non-mechanical grounding inherits any mechanical law beyond the structural form stated in Hypothesis 6.1.

11 Cognitive and institutional groundings

In cognition, Narrative-supporting relations may include expectations, habits, motor programmes, autobiographical commitments, causal explanations, or identity-bearing relations. Predictive coding provides an established example in which persistent higher-level expectations shape subsequent interpretation [7]; schema theory supplies a motor-control example in which learned generalized structures condition future movement [8]; and formal belief revision provides a model of dependency-sensitive restructuring [6].

In institutions, contracts, procedures, staffing, compatibility commitments, infrastructure, capital allocation, and reputation can carry Narrative. Path-dependence research provides direct precedent for the claim that historically accumulated structures can constrain later possibilities and become progressively costly to reverse [4, 5]. Routine output may therefore remain stable even when the institution's capacity to materially revise the structure carrying that routine has degraded.

These are groundings of the abstract structure, not claims that cognition or institutions are literally Newtonian particles.

12 Relationship to Narrativium

Narrative Momentum describes persistence. Narrativium describes capacity for deformation:

$$\boxed{\mathcal{N} = \text{operative history-bearing structure}, \qquad N = \text{reachable capacity to change structure.}}$$

Revision horizons produced by comparing current capacity with minimum revision cost are developed separately.

13 Positioning and prior art

The core ingredients of Narrative Momentum are individually old. Hysteresis theory formalises selective memory in dynamical systems [1, 2]. Path dependence and increasing returns formalise historically conditioned lock-in in economics and institutions [3–5]. AGM theory formalises dependency-sensitive belief revision [6]. Predictive coding and motor-schema theories formalise history-bearing internal structure affecting future processing and action [7, 8].

The claimed contribution is the reduction rather than the ingredients: these phenomena can be expressed as instances of one frame-relative pattern,

$$\boxed{\text{realised history} \to \text{operative present structure} \to \text{directional persistence of reachable futures.}}$$

Graphic Equalisation supplies the causal reason the pattern works: only present structure enters the next deterministic transition relation. Seldon manifolds supply the geometry in which that persistence is expressed. Narrative names the measurable higher-frame state that carries the surviving history.

14 Falsifiability commitments

1. **First-class Narrative.** Fails for a claimed grounding if Narrative cannot be measured as frame-level state and exists only as an after-the-fact interpretive label.
2. **Dimension/value typing.** Fails if absence of the Narrative relation must be represented as a numerical value of Narrative rather than as absence of the dimension.
3. **Discrete Narrative.** Fails if Narrative-dependent observable behaviour requires a non-discrete primitive that no dynamically quantised discrete representation can preserve.
4. **Historical sufficiency.** Fails if erased history with no present causal trace systematically changes future transition structure.
5. **Narrative Momentum (cost form).** Fails for a specified system class if a continuation requiring operative history-bearing structure to be altered carries no greater cost than one reusing it, absent any interaction that alters that structure. This is the primary commitment, and it is testable without a directional dimension. It also fails if history is found to remove continuations from the admissible set rather than to price them, which would place the effect in Γ (Remark 6).
6. **Narrative Momentum (directional form).** Fails for a frame instantiating a directional dimension if operative Narrative-supporting structure and its represented direction persist while the associated directional tendency repeatedly vanishes or reverses without a relevant interaction. Failure here without failure of the restriction form indicts the frame's directional instrumentation rather than the law.
7. **Structural lock-in.** Fails if increasing the structure that must be revised systematically lowers minimum revision cost after relevant capacity and geometry are controlled.
8. **History-sensitive projection.** Fails if histories that produce different operative present relations never produce different reachable projections in the claimed domain; conversely, the hypothesis is misapplied if different histories with identical causally complete present state are claimed to produce different deterministic successors.

15 Conclusion

Narrative is not a metaphor smuggled into dynamics. It is first-class state in the frame that contains the relations from which it is measured:

$$\boxed{\text{collapse} \to \text{realised history} \to \mathcal{N} \to \text{directional persistence in reachable geometry.}}$$

Momentum no longer requires a magical bridge. Narrative is already dynamical state in the higher frame; Narrative momentum is the directional persistence produced when that state participates in subsequent reachable geometry.

The constituent ideas have long precedents. The proposed novelty is that the same minimal recursive structure accounts for their common form without making Narrative a primitive of every graph or treating history as a force that acts from the past without surviving present structure.

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